

Chat Link: https://parley.mit.edu/share/bdUfy-qnaHaw8R8VQ_Tc2

NEEDS TO BE FIXED

****HAS HISTORIC DATA POINT STARTING HERE (PRODUCTS BEFORE NOW DO NOT YET HAVE HISTORIC POINT)****

```
global_data = {
  # --- ATTRIBUTE NAMES ---
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "mass",
  "attribute_4": "cross_sectional_area",
  "attribute_5": "connection_reliability",

  # --- DIRECTIONS ---
  "direction_attribute_1": "negative", # lower cost is better
  "direction_attribute_2": "positive", # higher copper history favors copper product
  "direction_attribute_3": "negative", # lower mass is better
  "direction_attribute_4": "negative", # smaller conductor area = more compact = better
  "direction_attribute_5": "positive", # higher reliability is better

  # --- ATTRIBUTE BOUNDS (robust across future datasets) ---

  # attribute_1: total cost ($USD)
  # Low end: ultra-low-cost region, small aluminum winding, minimal labor
  # High end: heavy-duty truck copper winding, premium labor market (e.g., Switzerland)
  "attribute_1_min": 1.00,
  "attribute_1_max": 40.00,

  # attribute_2: material continuity — hard binary scale, already at absolute limits
  "attribute_2_min": 0,
  "attribute_2_max": 1,

  # attribute_3: mass (kg)
  # Low end: motorcycle/micro-vehicle alternator winding
  # High end: heavy-duty commercial vehicle alternator winding
  "attribute_3_min": 0.15,
  "attribute_3_max": 1.50,

  # attribute_4: cross-sectional area (mm²)
  # Low end: very compact, high-efficiency copper winding in a premium small alternator
  # High end: large-gauge aluminum conductor in a high-output truck alternator
  "attribute_4_min": 50,
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"attribute_4_max": 500,

# attribute_5: connection reliability — normalized 0–1 scale, already at absolute limits
"attribute_5_min": 0.0,
"attribute_5_max": 1.0,

# --- MATERIAL PRICES ---
"copper_price_per_ton": 9500,
"aluminum_price_per_ton": 2500,
"historic_copper_price_per_ton": 8500,    # 2024
"historic_aluminum_price_per_ton": 2000,  # 2024
"copper_price_per_kg": 9.500,    # fixed global Cu price
"aluminum_price_per_kg": 2.500,  # fixed global Al price
}

```

```

regional_data = [
    {
        "region": "United States",
        "copper_product_market_share": 0.87, # UPDATED — slightly lower than
        "aluminum_product_market_share": 0.13, # India; capable of Al adoption
        "historic_copper_product_market_share": 0.91, # UPDATED
        "historic_aluminum_product_market_share": 0.09,
        "weight_attribute_1": 0.25,
        "weight_attribute_2": 0.20,
        "weight_attribute_3": 0.15,
        "weight_attribute_4": 0.20,
        "weight_attribute_5": 0.20,
    },
    {
        "region": "China",
        "copper_product_market_share": 0.80,
        "aluminum_product_market_share": 0.20,
        "historic_copper_product_market_share": 0.85,
        "historic_aluminum_product_market_share": 0.15,
        "weight_attribute_1": 0.35,
        "weight_attribute_2": 0.15,
        "weight_attribute_3": 0.20,
        "weight_attribute_4": 0.15,
        "weight_attribute_5": 0.15,
    },
    {
        "region": "India",
        "copper_product_market_share": 0.90, # UPDATED — highest Cu share;

```

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    "aluminum_product_market_share": 0.10, # harsh env + supply chain lock-in
    "historic_copper_product_market_share": 0.94, # UPDATED
    "historic_aluminum_product_market_share": 0.06,
    "weight_attribute_1": 0.40,
    "weight_attribute_2": 0.20,
    "weight_attribute_3": 0.15,
    "weight_attribute_4": 0.10,
    "weight_attribute_5": 0.15,
  },
  {
    "region": "Europe",
    "copper_product_market_share": 0.82,
    "aluminum_product_market_share": 0.18,
    "historic_copper_product_market_share": 0.87,
    "historic_aluminum_product_market_share": 0.13,
    "weight_attribute_1": 0.15,
    "weight_attribute_2": 0.20,
    "weight_attribute_3": 0.30,
    "weight_attribute_4": 0.20,
    "weight_attribute_5": 0.15,
  },
]

```

```
product_data = [
```

```

# =====
# UNITED STATES
# =====
{
  "region": "United States",
  "dominant material": "cu",
  "product": "ICE Alternator Winding",
  "copper_kg": 0.60,
  "aluminum_kg": 0.05,
  "total_mass_kg": 0.70,
  "non_material_cost_per_unit": 12.00,
  "attribute_2_value": 1, # material_continuity: full copper history
  "attribute_3_value": 0.70, # mass (kg)
  "attribute_4_value": 150, # cross-sectional area (mm²)
  "attribute_5_value": 0.90, # connection reliability
},
{

```

```

    "region": "United States",
    "dominant material": "al",
    "product": "ICE Alternator Winding",
    "copper_kg": 0.05,
    "aluminum_kg": 0.30,
    "total_mass_kg": 0.40,
    "non_material_cost_per_unit": 12.00,
    "attribute_2_value": 0,    # material_continuity: no copper history
    "attribute_3_value": 0.40, # mass (kg)
    "attribute_4_value": 246,  # cross-sectional area (mm²)
    "attribute_5_value": 0.65, # connection reliability
},

# =====
# CHINA
# =====
{
    "region": "China",
    "dominant material": "cu",
    "product": "ICE Alternator Winding",
    "copper_kg": 0.60,
    "aluminum_kg": 0.05,
    "total_mass_kg": 0.70,
    "non_material_cost_per_unit": 5.00,
    "attribute_2_value": 1,    # material_continuity: full copper history
    "attribute_3_value": 0.70, # mass (kg)
    "attribute_4_value": 150,  # cross-sectional area (mm²)
    "attribute_5_value": 0.90, # connection reliability
},
{
    "region": "China",
    "dominant material": "al",
    "product": "ICE Alternator Winding",
    "copper_kg": 0.05,
    "aluminum_kg": 0.30,
    "total_mass_kg": 0.40,
    "non_material_cost_per_unit": 5.00,
    "attribute_2_value": 0,    # material_continuity: no copper history
    "attribute_3_value": 0.40, # mass (kg)
    "attribute_4_value": 246,  # cross-sectional area (mm²)
    "attribute_5_value": 0.65, # connection reliability
},

# =====

```

INDIA

=====

```
{
  "region": "India",
  "dominant material": "cu",
  "product": "ICE Alternator Winding",
  "copper_kg": 0.60,
  "aluminum_kg": 0.05,
  "total_mass_kg": 0.70,
  "non_material_cost_per_unit": 4.00,
  "attribute_2_value": 1,    # material_continuity: full copper history
  "attribute_3_value": 0.70, # mass (kg)
  "attribute_4_value": 150,  # cross-sectional area (mm²)
  "attribute_5_value": 0.90, # connection reliability
},
```

```
{
  "region": "India",
  "dominant material": "al",
  "product": "ICE Alternator Winding",
  "copper_kg": 0.05,
  "aluminum_kg": 0.30,
  "total_mass_kg": 0.40,
  "non_material_cost_per_unit": 4.00,
  "attribute_2_value": 0,    # material_continuity: no copper history
  "attribute_3_value": 0.40, # mass (kg)
  "attribute_4_value": 246,  # cross-sectional area (mm²)
  "attribute_5_value": 0.65, # connection reliability
},
```

=====

EUROPE

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```
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "ICE Alternator Winding",
  "copper_kg": 0.60,
  "aluminum_kg": 0.05,
  "total_mass_kg": 0.70,
  "non_material_cost_per_unit": 14.00,
  "attribute_2_value": 1,    # material_continuity: full copper history
  "attribute_3_value": 0.70, # mass (kg)
  "attribute_4_value": 150,  # cross-sectional area (mm²)
  "attribute_5_value": 0.90, # connection reliability
}
```

```
},  
{  
  "region": "Europe",  
  "dominant material": "al",  
  "product": "ICE Alternator Winding",  
  "copper_kg": 0.05,  
  "aluminum_kg": 0.30,  
  "total_mass_kg": 0.40,  
  "non_material_cost_per_unit": 14.00,  
  "attribute_2_value": 0,    # material_continuity: no copper history  
  "attribute_3_value": 0.40, # mass (kg)  
  "attribute_4_value": 246,  # cross-sectional area (mm2)  
  "attribute_5_value": 0.65, # connection reliability  
},  
]
```